



Bike Sharing Demand Analysis



Overview

This project focuses on analyzing a **bike sharing dataset** to understand the factors affecting bike rental demand. The goal is to explore patterns in usage and identify key variables that influence the number of bike rentals.

This case study is widely used in data analytics and machine learning to demonstrate **EDA, feature engineering, and regression modeling**.



Problem Statement

Bike-sharing systems generate large amounts of data that can be used to:

- Predict bike rental demand
- Optimize bike availability
- Improve business decisions

The objective is to analyze historical data and build insights into demand patterns based on environmental and seasonal factors.



Dataset Description

The dataset includes hourly/daily records of bike rentals with features such as:

- Date & time
 - Season (Spring, Summer, Fall, Winter)
 - Weather conditions
 - Temperature & humidity
 - Windspeed
 - Working day / Holiday
 - Total bike rentals (target variable)
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Exploratory Data Analysis (EDA)

Data Cleaning

- Handled missing values
- Removed irrelevant columns
- Converted categorical variables

- Checked for outliers

Univariate Analysis

- Distribution of rentals across seasons
- Temperature and humidity trends
- Usage patterns across days

Bivariate Analysis

- Rentals vs temperature
- Rentals vs weather conditions
- Rentals vs working day / holiday

Multivariate Analysis

- Correlation heatmap
 - Feature interactions impacting demand
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Key Insights

- Bike rentals are higher during **favorable weather conditions**
 - Demand peaks during **working days and rush hours**
 - Temperature has a strong positive correlation with rentals
 - Extreme weather (rain, high humidity) reduces demand significantly
 - Seasonal trends show higher usage in summer and fall
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Modeling Approach (Optional)

- Linear Regression
 - Ridge / Lasso Regression
 - Decision Tree / Random Forest (if applied)
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Evaluation Metrics

- R^2 Score
 - RMSE (Root Mean Squared Error)
 - MAE (Mean Absolute Error)
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Tech Stack

- Python

- Pandas, NumPy
 - Matplotlib, Seaborn
 - Scikit-learn
 - Jupyter Notebook
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Conclusion

This analysis helps in understanding the demand patterns of bike-sharing systems and enables better planning and resource allocation. The insights can be used to improve operational efficiency and customer satisfaction.

Future Work

- Build advanced predictive models (XGBoost, LightGBM)
- Deploy demand prediction API
- Integrate real-time weather data for dynamic predictions